Principal Component Analysis (PCA) of Surface Contamination by TOF-SIMS

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***Abstract*— Static time of flight secondary ions mass spectrometry (TOF-SIMS) is a qualitative analysis technique with a high detection limit capable of achieving parts per million (ppm) levels [1] and sub monolayer surface detection. Due to its high sensitivity, the data obtained may contains more than hundreds of peaks including information about sample composition, chemical compound, fragments ions, sample purity, and etc. [2]. Therefore, TOF-SIMS data interpretation is complicated and time consuming especially for unknown contamination. These challenges could be overcome by Principal Component Analysis (PCA). PCA is one of a multivariate method used to identify mass peak spectral in various principal component loadings when processing large data sets such as TOF-SIMS data. This paper presents a study of surface contamination analysis using image based PCA to find out the parent ion of the contaminant.**

Keywords—TOF-SIMS, Imaging PCA, contamination

I. INTRODUCTION

TOF-SIMS is one of the surface analysis technique that is suitable in trace contamination identification. However, TOF-SIMS data interpretation is relatively complicated even with the simplest surface analysis [1]. The complexion includes interpreting spectra containing more than hundreds of peaks, interrelation between the peak intensities, over-lapping peaks, mass interference, matrix effect and etc. Furthermore, the complexion increase when one analyses TOF-SIMS data from TOF-SIMS imaging as each pixel of a data image contains information of the entire mass spectrum, i.e. 3584 spectra in a 128 x 128 image or 65536 spectra in a 256x256 image [2].

TOF-SIMS data interpretation commonly involve manual peaks selection that is dependent on the analyst's understanding on the sample, or derived from an averaged spectrum. Moreover, one could loss essential information if certain peaks were not identified [3] due to the drawback that co-existing materials, mass interference or surface density could affect data extraction.

Multivariate Analysis (MVA) is very useful in data process especially for large data sets. There are a few typical MVA methods namely, Principal Component Analysis (PCA), Factor Analysis (FA), Discriminant Analysis (DA), Multivariate Curve Resolution (MCR), Maximum Autocorrelation Factors (MAF) etc. Among all these methods, PCA is one of the most commonly used technique in TOF-SIMS data interpretation [2-5]. The interpretation depends on the covariance between all the variables of the analysis. This method could remove the analyst bias in the interpretation process and essentially provide useful information such as potential characteristic or signature peaks of the contamination.

In this paper, the study of data interpretation using PCA method will be based on the morphology characterization of the contamination on the surface of the sample or the chemical distinction of the contamination from the sample surrounding. The identification of the PCA is interlinked between TOF-SIMS mapping images and mass spectral. The factor loadings presented in the image scores from the principal component (PC) differs across a similar analysis area. The PCA will be used to identify the spectral region of the contamination from the surrounding within the imaged area.

II. EXPERIMENTAL

The analysis was acquired by TOF-SIMS[5] instrument (ION TOF GmbH, Munster, Germany) with the parameter as in Table I.

TABLE I. ANALYSIS ACQUIRED PARAMETER

Parameter	
Species	Bi+
Analysis area (um x um)	500 x 500
Cycle time (us)	100
Energy (eV)	250000

From the TOF-SIMS mapping image shown in Figure 1, the brighter regions indicate traces of contamination present on the analyzed sample.

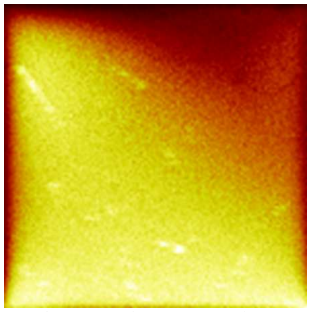


Fig. 1. Total counts of TOF-SIMS ions mapping distribution

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A. *Data Processing*

With the intention to obtain a thorough set of data, the automated peak selection from the software SurfaceLab 6.3 (IONTOF, Munster, Germany) is utilized to overcome loss of information. The mass peaks were selected by the automated peak search routine with the following conditions: Boundary conditions: m/z = 1-884, signal intensity > 500-1000 counts and Signal to noise (SNR) > 3.0. Example of automated peak search shown in Figure 2. The value input in SNR peaks during selection is useful as it increases the likelihood to identify signature peaks that is crucial to identify a contamination in some cases [4].

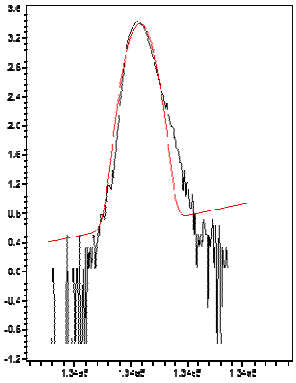


Fig. 2. Illustration of mass peak spectra using automated peak search function

III. RESULT AND DISCUSSION

The overall spectra of raw data for both positive and negative polarity consist of more than hundred peaks seen from Figure 3 and 4. Data interpretation from raw data is challenging, which includes difficulty to distinguish mass peaks and identification of signature peaks that are essential in determining chemical or contamination traces.

In order to extract more data information, PCA technique is employed in this study. PCA provides an outline information of the contamination from the surrounding surface. The factor loadings presented in the image scores from the PC differs across a similar analysis area. Theoretically, the factor loading of PC1 has the strongest influence over the entire mass range due to a phenomenon known as topographical effects [6] which potentially affects all or most of the mass peak intensities uniformly. As all mass fragments are affected under this influence, PC1 accounts for the majority of mass variance.

The loading of PC1 for both positive and negative polarity are shown in Figure 5 and 6, respectively. With respect to the sample used in this analysis, PC1 loading is more prominent in the positive polarity as compared to the negative polarity.

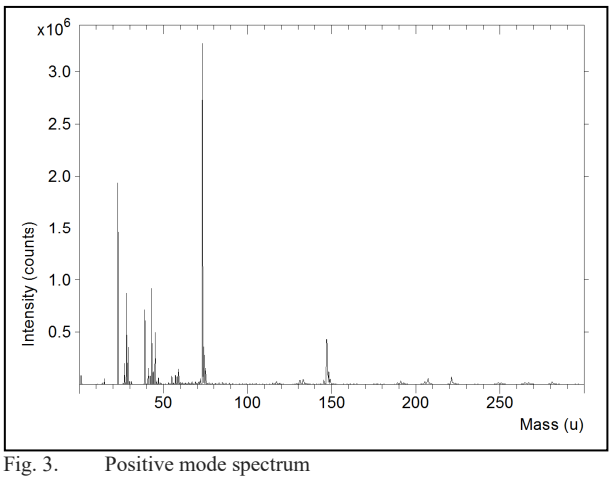


Fig. 3. Positive mode spectrum

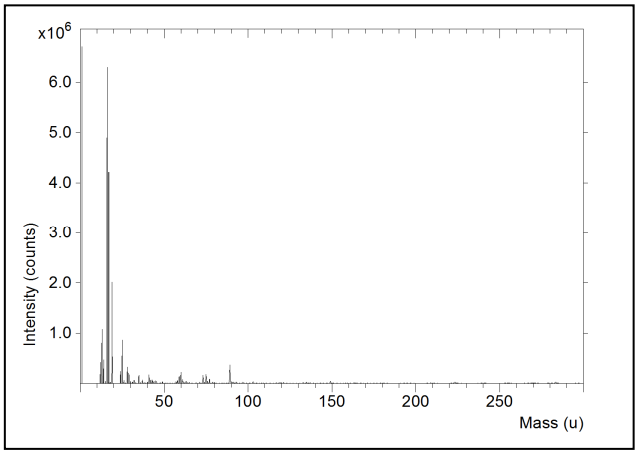


Fig. 4. Negative mode spectrum

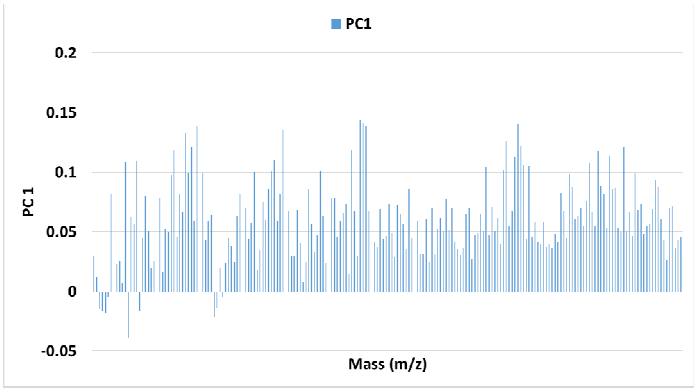


Fig. 5. PC1 loading in positive polarity

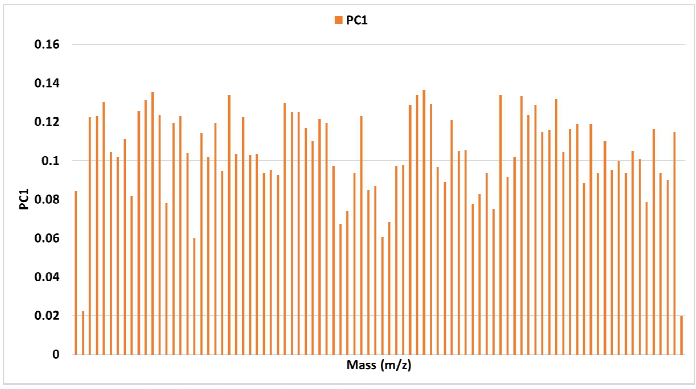


Fig. 6. PC1 loading in negative polarity

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To account for the mass variance present in PC1, the subsequent PC loadings with reduced noise and better contrast i.e., PC2 (Figure 7) in positive polarity and PC3 (Figure 8) in negative polarity were found to be useful in this study. As a result, PC2 and PC3 were further studied and discuss in the next section.

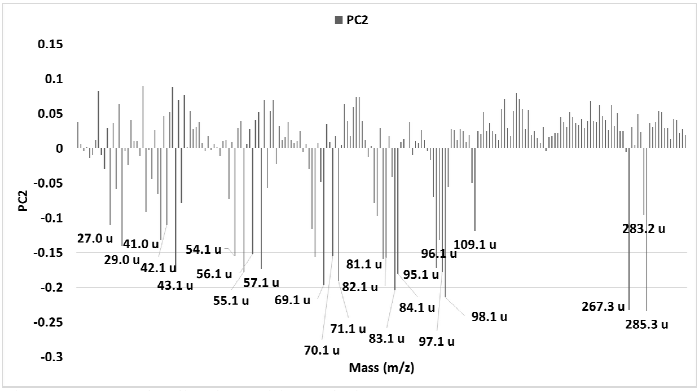


Fig. 7. PC2 loading in positive polarity

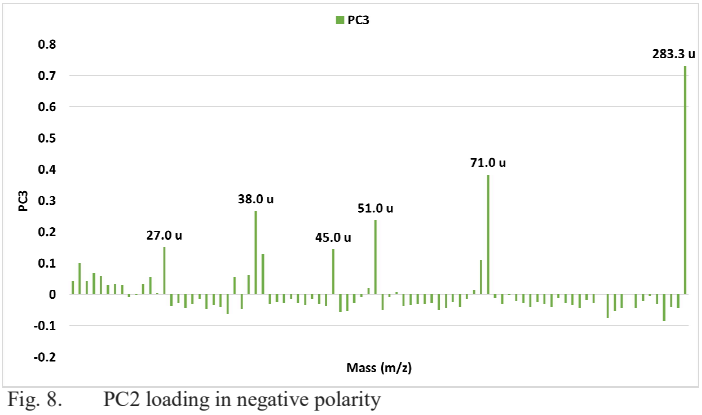


Fig. 8. PC2 loading in negative polarity

From Figure 7, the peaks present in the negative region of the PC2 loading represents the contamination peaks fragment corresponding to the ions image in Figure 9. Mass peaks fragments eg. 27.0u, 29.0u, 41.0u, 42.1u, 43.1u, etc. present in PC2 loading is validated using the mapping images corresponding to the mass peak fragments. From Figure 9, the mapping images with brighter regions corresponded with the distinctive masses obtained from PC2. In other words, the PCA technique is useful in identifying or differentiating the presence of contamination from its surface surrounding as the intensity of mass peaks from its respective ion mapping image is strongly correlated to the mass distinct in their respective PCA loading spectrum.

Similarly, the mass peaks fragments in the negative polarity are more observable in the positive region of PC3 loading as seen from Figure 8, corresponding to its ion mapping image shown in Figure 10.

From the correlation study between the ion image mapping and PC loadings, the mass peak fragments suggest the presence of contamination which consist of aliphatic hydrocarbon (CxHy) and oxygen contain hydrocarbon (CxHyO). The positive and negative chemical compounds present in this contamination are tabulated in Table 2 and 3, respectively.

The distinctive peaks obtained from PC2 loading with high masses 267.3u and 285.3u in positive polarity (Figure 9) and 283.3u in negative polarity (Figure 10) suggest that these peaks are possible to be signature peaks contributing to the contamination. The mass of the identified signature peaks corresponds to that of long chain fatty stearic acid [7].

TABLE II. LIST OF DISTINCT POSITIVE IONS FRAGMENTS OBTAINED FROM PC LOADING.

Mass (m/z)	Ions
27.0	C_2H_3+
29.0	C_2H_5+
41.0	C_3H_5+
42.1	C_3H_6+
43.1	C_3H_7+
54.1	C_4H_6+
55.1	C_4H_7+
56.1	C_4H_8+
57.1	C_4H_9+
69.1	C_5H_9+
70.1	C_5H_10+
71.1	C_5H_11+
81.1	C_6H_9+
82.1	C_6H_10+
83.1	C_6H_11+
95.1	C_7H_11+
97.1	C_7H_13+
109.1	C_8H_13+
84.1	C_5H_8O+
96.1	C_6H_8O+
98.1	C_6H_10O+
267.3	C_18H_35O+
283.2	C_18H_35O_2+
285.3	C_18H_37O_2+

TABLE III. LIST OF DISTINCT NEGATIVE IONS FRAGMENTS OBTAINED FROM PC LOADING.

Mass (m/z)	Ions
27.0	C_2H_3-
38.0	C_3H_2-
45.0	CHO_2-
51.0	C_4H_3-
71.0	C_3H_3O_2-
283.3	C_18H_35O_2-

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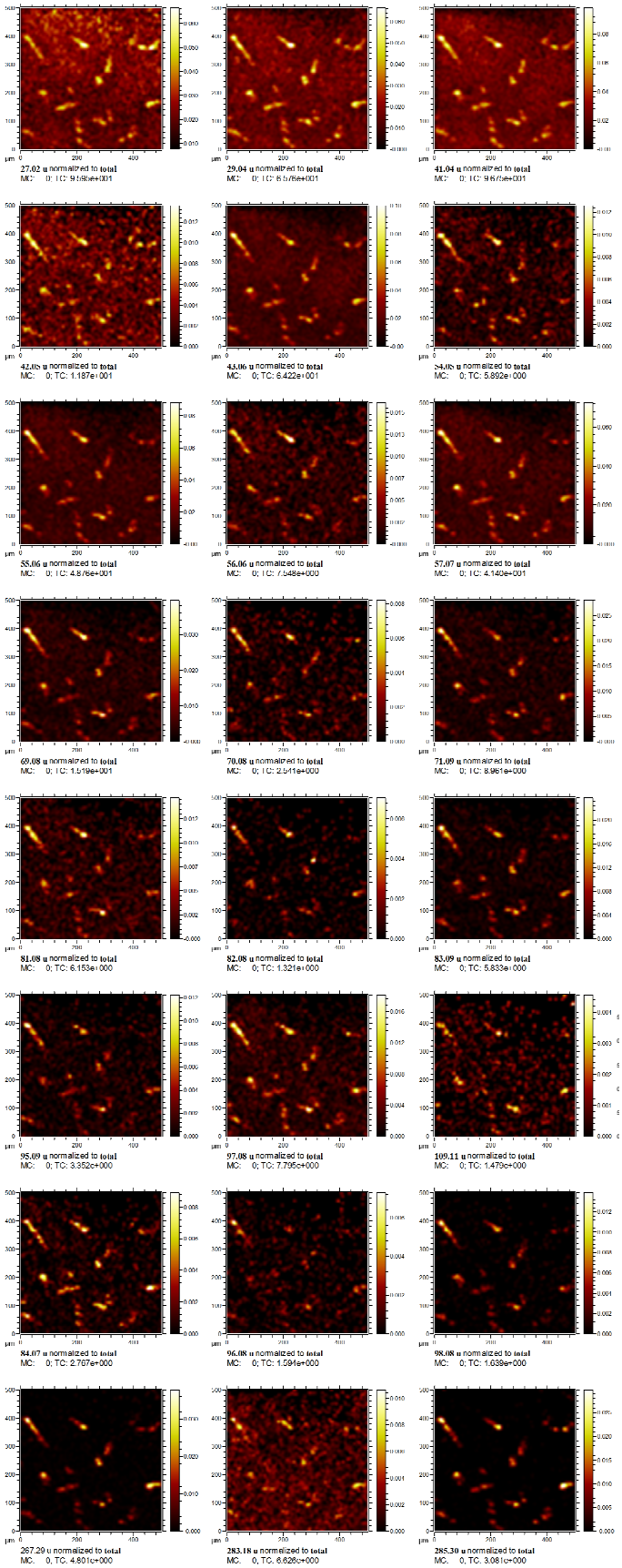


Fig. 9. Positive ions mapping images

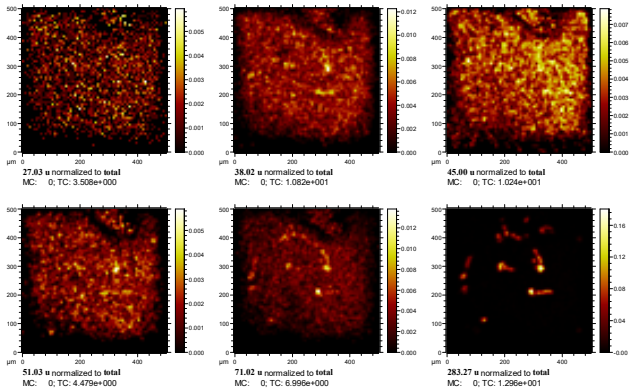


Fig. 10. Negative ions mapping images

IV. CONCLUSION

PCA is one of the MVA that is useful in TOF-SIMS data interpretation. The mass peaks obtained from the PCA loadings correspond to the ions mapping image which is essentially useful in identifying the signature peaks and chemical composition of the unknown contamination from the surrounding. As in this study, the signature peaks of 267.3u (C_18H_35O+), 285.3u (C_18H_37O_2+) and 283.3u (C_18H_35O_2-) were identify corresponding to the long chain fatty of stearic acid [7].

V. ACKNOWLEDGMENT

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